

24k-0559

BSCS 4-H

LAB TASKS 1 AI LAB

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TASK1:

CODE:

```
num1 = int(input("Enter first number "))
```

```
num2 = int(input("Enter second number "))
```

```
print(num1 + num2)
```

```
print(type(num1))
```

```
print(type(num2))
```

OUTPUT:

```
[33]: #Task1
num1 = int(input("Enter first number "))
num2 = int(input("Enter second number "))

print(num1 + num2)
print(type(num1))
print(type(num2))

Enter first number 2
Enter second number 3
5
<class 'int'>
<class 'int'>
```

TASK2:

CODE:

```
name = input("Enter Name ")
print(name[0])
print(name[int(len(name)/2)])
print(name[len(name)-1])
print(len(name))
```

OUTPUT:

```
<class int >  
<class 'int'>
```

```
[34]: #Task2  
  
name = input("Enter Name ")  
print(name[0])  
print(name[int(len(name)/2)])  
print(name[len(name)-1])  
print(len(name))
```

```
Enter Name  bazil  
b  
z  
l  
5
```

```
[16]: #Task 3
```

TASK3:

CODE:

```
list = [1,8.9,"BAZ",'B']
```

```
list[2]="bcv"
```

```
list.append(78)
```

```
list.pop()
```

```
print(list)
```

OUTPUT:

```
[35]: #Task 3
list = [1,8.9,"BAZ",'B']
list[2]="bcv"
list.append(78)
list.pop()
print(list)

[1, 8.9, 'bcv', 'B']
```

TASK4:

CODE:

mytup = (1,2,3,4,5)

sum =0

for x in range(len(mytup)):

print(mytup[x])

sum = sum + mytup[x]

print("The Sum Is : " , sum)

OUTPUT:

```
[22]: #Task 4
      mytup = (1,2,3,4,5)
      sum =0
      for x in range(len(mytup)):
          print(mytup[x])
          sum = sum + mytup[x]

      print("The Sum Is : " , sum)

1
2
3
4
5
The Sum Is : 15
```

TASK5:

CODE:

listnum = []

for x in range(10):

num = int(input("Enter Number "))

listnum.append(num)

set_list = set(listnum)

print(listnum)

print(set_list)

removed = len(listnum) - len(set_list)

print("Number of duplicatesremoved:", removed)

OUTPUT:

```
[2]: #Task 5

listnum = []

for x in range(10):
    num = int(input("Enter Number "))
    listnum.append(num)

set_list = set(listnum)
print(listnum)
print(set_list)
removed = len(listnum) - len(set_list)
print("Number of duplicatesremoved:", removed)

Enter Number 1
Enter Number 2
Enter Number 3
Enter Number 4
Enter Number 5
Enter Number 6
Enter Number 7
Enter Number 8
Enter Number 0
Enter Number 8
[1, 2, 3, 4, 5, 6, 7, 8, 0, 8]
{0, 1, 2, 3, 4, 5, 6, 7, 8}
Number of duplicatesremoved: 1
```

TASK6:

CODE:

```
dictiona = {"Name" : "Qadir","Marks" : 90,"Grade" : " "}
```

```
if(dictiona["Marks"] >=90):  
    dictiona["Grade"] = "A+"  
elif (dictiona["Marks"] >=80 and dictiona["Marks"] <90):  
    dictiona["Grade"] = "A"  
elif (dictiona["Marks"] >=70 and dictiona["Marks"] <80):  
    dictiona["Grade"] = "B"  
elif (dictiona["Marks"] >=60 and dictiona["Marks"] <70):  
    dictiona["Grade"] = "C"  
else:  
    dictiona["Grade"] = "F"  
  
print(dictiona)
```

OUTPUT:

```
]: #Task 6
dictiona = {"Name" : "Qadir", "Marks" : 90, "Grade" : " "}

if(dictiona["Marks"] >=90):
    dictiona["Grade"] = "A+"
elif (dictiona["Marks"] >=80 and dictiona["Marks"] <90):
    dictiona["Grade"] = "A"
elif (dictiona["Marks"] >=70 and dictiona["Marks"] <80):
    dictiona["Grade"] = "B"
elif (dictiona["Marks"] >=60 and dictiona["Marks"] <70):
    dictiona["Grade"] = "C"
else:
    dictiona["Grade"] = "F"

print(dictiona)

{'Name': 'Qadir', 'Marks': 90, 'Grade': 'A+'}
```

TASK7:

CODE:

```
num1 = int(input("Enter Number " ))
```

```
if(num1 <= 1):
```

```
    print("Number Is Not Prime")
```

```
else:
```

```
    flag = False
```

```
    for x in range(2,num1):
```

```
        if(num1 % x == 0):
```

flag = True

if(flag == False):

print("Number is prime")

else:

print("Number is not prime")

OUTPUT:

```
[27]: #Task 7
num1 = int(input("Enter Number " ))

if(num1 <= 1):
    print("Number Is Not Prime")
else:
    flag = False
    for x in range(2,num1):

        if(num1 % x == 0):
            flag = True

    if(flag == False):
        print("Number is prime")
    else:
        print("Number is not prime")
```

```
Enter Number 4
Number is not prime
```

TASK8:

CODE:

```
count =0
for x in range(1,51):
    if(x % 2 == 0):
        print("Even Number ",x)
        count = count + 1

print("Total Even Numbers : ",count)
```

OUTPUT:

```
[28]: #TASK 8
count =0
for x in range(1,51):
    if(x % 2 == 0):
        print("Even Number ",x)
        count = count + 1

print("Total Even Numbers : ",count)
```

```
Even Number  2
Even Number  4
Even Number  6
Even Number  8
Even Number 10
Even Number 12
Even Number 14
Even Number 16
Even Number 18
Even Number 20
Even Number 22
Even Number 24
Even Number 26
Even Number 28
Even Number 30
Even Number 32
Even Number 34
Even Number 36
Even Number 38
Even Number 40
Even Number 42
Even Number 44
Even Number 46
Even Number 48
Even Number 50
Total Even Numbers :  25
```

TASK9

CODE:

```
multiplier = 1
num = int(input("Enter A number " ))
while (multiplier <= 10):
    print(multiplier * num)
    multiplier = multiplier+ 1
```

OUTPUT:

```
[31]: #Task 9

multiplier = 1
num = int(input("Enter A number " ))
while (multiplier <= 10):
    print(multiplier * num)
    multiplier = multiplier+ 1
```

```
Enter A number 2
2
4
6
8
10
12
14
16
18
20
```

TASK10:

Code:

```
def details(*numbers):
```

```
    return max(numbers)
```

```
print(details(10, 25, 3, 99, 45))
```

```
def printdetails(**n):
```

```
    for key, value in n.items():
```

```
        print(key, ":", value)
```

```
printdetails(name="Qadir", marks=90, grade="A+")
```

Output:

20

```
[32]: #Task 10
def details(*numbers):
    return max(numbers)

print(details(10, 25, 3, 99, 45))

def printdetails(**n):

    for key, value in n.items():
        print(key, ":", value)

printdetails(name="Qadir", marks=90, grade="A+")
```

```
99
name : Qadir
marks : 90
grade : A+
```

```
[ ]:
```